Master Exam 2018

Problem 1. Let $X_1, \ldots, X_n$ be i.i.d. from a uniform distribution on $[0, \theta]$.

(1) Find the maximum likelihood estimator $\hat{\theta}$ of θ.

(2) Show that $\hat{\theta}/\theta$ is a pivotal quantity and use it to set a $1-\alpha$ confidence interval for θ.

Solution. (1) The likelihood function is $\theta^{-n}\mathbf{1}_{X_{(n)}\leq\theta}$ so is maximized at $\hat{\theta} = X_{(n)}$.

(2) For any $x \in (0,1)$,

$$P(\hat{\theta}/\theta \leq x) = P(X_{(n)} \leq x\theta) = \prod_{i=1}^{n} P(X_i \leq x\theta) = x^n,$$

so $\hat{\theta}/\theta$ is pivotal. The $(\alpha/2)$-th and $(1-\alpha/2)$-th quantiles for this distribution are $(\alpha/2)^{1/n}$ and $(1-\alpha/2)^{1/n}$. So

$$\begin{aligned} 1-\alpha &= P\big[(\alpha/2)^{1/n} \leq \hat{\theta}/\theta \leq (1-\alpha/2)^{1/n}\big] \\ &= P\left[\theta \in \left[\frac{\hat{\theta}}{(1-\alpha/2)^{1/n}}, \frac{\hat{\theta}}{(\alpha/2)^{1/n}}\right]\right]. \end{aligned}$$

Hence

$$\left[\frac{\hat{\theta}}{(1-\alpha/2)^{1/n}}, \frac{\hat{\theta}}{(\alpha/2)^{1/n}}\right]$$

is the desired $1-\alpha$ confidence interval for θ.

Problem 2. Let $Y_1, \ldots, Y_n$ be a random sample from an absolutely continuous distribution with density

$$f_\theta(x) = \frac{2x}{\theta^2}\mathbf{1}_{(0,\theta)}.$$

(1) Find a one-dimensional sufficient statistic T.

(2) Determine the density of T.

Solution. (1) Let $T(x) = \max\{x_1, \ldots, x_n\}$, and $M(x) = \min\{x_1, \ldots, x_n\}$. Then the joint density will be positive if and only if $M(x) > 0$ and $T(x) < \theta$. So the joint density can be written as

$$\prod_{i=1}^{n}(2x_i)\,\mathbf{1}_{M(x)>0}\mathbf{1}_{T(x)<\theta}/\theta^{2n}.$$

$T = T(X)$ is sufficient by the factorization theorem.

(2) For $t \in (0, \theta)$,

$$P\left(X_i \leq t\right) = \int_0^t \frac{2x}{\theta^2} \mathrm{d}x = \frac{t^2}{\theta^2}.$$

So

$$\begin{aligned} P\left(T \leq t\right) &= P\left(X_1 \leq t, \ldots, X_n \leq t\right) \\ &= P\left(X_1 \leq t\right) \times \cdots \times P\left(X_n \leq t\right) = \frac{t^{2n}}{\theta^{2n}}. \end{aligned}$$

Taking the derivative of this with respect to t, T has density

$$t \mapsto 2nt^{2n-1}/\theta^{2n}, t \in \left(0, \theta\right).$$